



Nazmi Yücel Çan

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in model development, optimization and edge AI deployment. Reduced SLM model size 5x (500MB → 100MB) via pruning, 8-bit LoRA, and INT8 quantization. Experienced in real-time computer vision pipelines (YOLOv8-seg) and embedded ML (TinyML/Arduino). Co-authored a deep learning research paper published in El-Cezeri Journal (2026). Top 10 finalist at the Pupilica AI Hackathon. I share my technical insights and project deep-dives on my personal blog.

SKILLS

Programming Languages

Python (Advanced), SQL, Bash/Shell Scripting

ML & AI Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn,
Hugging Face (LoRA/PEFT), ONNX Runtime

Core Domains

Computer Vision (OpenCV, YOLOv8), NLP (Transformers), Data Preprocessing & Evaluation, Model Optimization (Quantization & Pruning), MLOps Fundamentals

Tools & Platforms

Git/GitHub, Linux (Ubuntu/Debian), PyQt5, Google Gemini API

Languages: Turkish (Native), English (Upper-Intermediate — Professional working proficiency)

EXPERIENCE

Machine Learning Intern at Deep Think Technology

02/2026 – 06/2026

- Developed an "Apple Detection and Quality Analysis System" using YOLOv8-seg for real-time fruit detection and CIEDE2000 color quality analysis.
- Designed and implemented a desktop GUI with PyQt5, integrating camera SDK, ML inference, and physical dimension measurement modules.
- Trained and deployed computer vision models on embedded devices for real-time edge AI inference.
- Implemented data preprocessing and model evaluation workflows for edge computing hardware, addressing real-world latency and resource constraints.

PROJECTS

mT5 Bidirectional EN-TR Machine Translation (ONNX INT8 & ORT Mobile)

Python, PyTorch, Hugging Face PEFT (LoRA), ONNX Runtime, INT8 Quantization, SLM

- Fine-tuned an mT5-small model with 8-bit LoRA for bidirectional English-Turkish translation (EN<->TR). Achieved a 5x model size reduction (500MB → 100MB) and optimized performance for resource-constrained environments by applying model pruning, ONNX compilation, and INT8 quantization for mobile deployment via ONNX Runtime (ORT Mobile).

Kaggle

Apple Detection and Quality Analysis System

Python, PyQt5, OpenCV, YOLOv8, MVS SDK

- An industrial computer vision application featuring real-time object detection, physical dimension measurement, and CIEDE2000 color quality analysis. Focused on system integration, designing the application architecture, and orchestrating the end-to-end pipeline—integrating the camera SDK, real-time YOLOv8-seg model inference, and measurement modules into a unified desktop interface.

GitHub

Arduino PC System Monitor

Python, Arduino (C++), PlatformIO, TinyML (SVM), scikit-learn

- A TinyML-powered system that monitors PC hardware resources (CPU, RAM, GPU, VRAM) and automatically optimizes power consumption. The system extracts hardware telemetry in real-time, executing an on-device SVM classification model on an Arduino Leonardo to determine and adjust the system's power state.

GitHub

Zettelkasten AI Notes (Pupilica AI Hackathon — Top 10 Finalist)

Python, Flet, SQLite, Google Gemini AI (google-genai), pypdf

- A knowledge management desktop application built around the Zettelkasten method. Features an LLM-powered structured information extraction and parsing pipeline that processes PDF documents via the Google Gemini API to extract atomic notes, automatically formats them using customized templates, and orchestrates links between notes. Modernized by migrating the UI to Flet, integrating flet.canvas for mind map rendering, and upgrading to the google-genai SDK. All processed content is stored in a local SQLite database.

GitHub

PUBLICATIONS

Deep Learning Models Integrating Attention Mechanisms For Military Camouflaged Object Detection — Nilgün Şengöz, Gül Karaman, Mert Samet Çeliker, Nazmi Yücel Çan
El-Cezeri Fen ve Mühendislik Dergisi (2026) — <https://doi.org/10.31202/ecjse.1747013>

EDUCATION

Bachelor of Science in Information Systems and Technologies 09/2022 – 06/2026
Burdur Mehmet Akif Ersoy University
GPA: 3.17/4.00

REFERENCES

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